



Introdução ao Biopython

Juliana Felix

DIA 01

```
>>> pip install biopython
```

Objetivos do Dia

- Entender o que é o Biopython e as principais aplicações;
- Conhecer o objeto do tipo Seq e os métodos básicos de manipulação;
- Realizar a transcrição, retrotranscrição, reverse complemento e tradução de sequências;
- Conhecer o objeto do tipo SeqRecord e sua aplicação na anotação de sequências.

```
>>> pip install biopython
```




O que é o Biopython?

- Conjunto de ferramentas de biologia molecular computacional
- Open source
- Escrita em Python

- Pode ser usado para manipulação de sequências, anotação, biologia estrutural e fácil acesso aos principais bancos de dados biológicos.

História

- **1999:** Início do desenvolvimento
- **2000:** Primeira versão
- **2002:** Primeira versão estável

Idealizado por **Jeff Chang** e **Andrew Dalke**, inspirados por Ewan Birney e pelo projeto BioPerl. Na sequência, **Brad Chapman** se uniu a eles e tornou-se um dos grandes colaboradores do projeto.

Desde **2012**, superou o BioPerl e passou a ser considerado o **Bio*Projeto mais popular** mantido pela **Open Bioinformatics Foundation**.

GenBank Files

.gb
.gbff
.gbk

```

LOCUS       NC_009418             4026 bp    DNA     linear     CON 16-JUL-2015
DEFINITION  Leishmania infantum 3PCH5 genome chromosome 34.
ACCESSION   NC_009418 REGION: 438960..442985
VERSION     NC_009418.2
DBLINK      BioProject: PRJNA101872
Assembly:  GCT_00000202875.2
KEYWORDS    RefSeq; complete genome.
SOURCE      Leishmania infantum 3PCH5
  ORGANISM  Leishmania infantum 3PCH5
            Eukaryota; Discoba; Euglenozoa; Kinetoplastes; Metakinetoplastina;
            Trypanosomatida; Trypanosomatidae; Leishmaninae; Leishmania.
REFERENCE   1
  AUTHORS   Rogers,M.B., Hilley,J.D., Dickens,N.J., Wilkes,J., Bates,P.A.,
            Depledge,D.P., Harris,D., Herzyk,P., Imamura,H., Otto,T.D.,
            Sanders,H., Seeger,K., Dujardin,J.C., Berriman,M., Smith,D.F.,
            Hertz-Fowler,C. and Mottram,J.C.
  TITLE     Chromosome and gene copy number variation allow major structural
            change between species and strains of Leishmania
  JOURNAL   Genome Res. 21 (12), 2129-2142 (2011)
  PUBMED   21938152
REFERENCE   2
  AUTHORS   Peacock,C.S., Seeger,K., Harris,D., Murphy,L., Ruiz,J.C.,
            Quail,M.A., Peters,N., Adlem,E., Tivey,A., Aslett,H., Kerhornou,A.,
            Ivens,A., Fraser,A., Rajandream,M.A., Carver,T., Norbertczak,H.,
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            Rutter,S., Squares,R., Whitehead,S., Rabinowitsch,E.,
            Arrowsmith,C., White,B., Thurston,S., Bringaard,F., Baldauf,S.L.,
            Faulconbridge,A., Jeffares,D., Depledge,D.P., Oyola,S.O.,
            Hilley,J.D., Brito,L.O., Tosi,L.R., Barrell,B., Cruz,A.K.,
            Mottram,J.C., Smith,D.F. and Berriman,M.
  TITLE     Comparative genomic analysis of three Leishmania species that cause
            diverse human disease
  JOURNAL   Nat. Genet. 39 (7), 839-847 (2007)
  PUBMED   17574675

```

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```

Thank You